Xiaojia Li

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EDUCATION

Carleton College **Northfield, MN**
Bachelor of Arts in Computer Science & Math | ***GPA: 3.98/4.00*** Jun 2026
Honors: Dean's List (2023 & 2024), Patricia V. Damon Scholarship (2025)

SKILLS

Focus Areas: Full-Stack Engineering, Generative AI, Human-AI Interaction
Programming Languages: C++, Python, Java, Kotlin, HTML, CSS, JavaScript, TypeScript, SQL, Scheme
Frameworks and Tools: Git, Docker, React, Flask, Svelte, Three.js, PostgreSQL, AWS, RESTful API, WebSocket, Linux OS, Raspberry Pi, ESP32
Methodologies: Object-Oriented Programming, Agile, Scrum, SDLC, Unit Testing, Prompt Engineering

PROFESSIONAL EXPERIENCES & PROJECTS

ByteDance **San Jose, CA**
Software Engineer Intern May 2025 – Aug 2025

- Developed a real-time interactive prototype for a novel **LLM-powered AI device** (under NDA), designing and implementing a responsive embedded system using ESP32, touchscreen, LEDs, and haptics
- Translated high-level interaction concepts from UI/UX and industrial design teams into synchronized feedback behaviors, collaborating across disciplines and showcased the system to ByteDance CEO during early product exploration
- Created and deployed styled **Three.js visualizations** with animated transitions to reflect AI status during voice-mode interaction (e.g. listening, thinking, responding); integrated with DouBao (an in-house speech-to-speech LLM) via FastAPI and WebSockets, and deployed on Render and Vercel

Lytup **Minneapolis, MN**
Software Engineer Intern Nov 2024 – Jan 2025

- Migrated and refactored 40+ pages and components from **Svelte 4** to **Svelte 5**, resolving 20+ deprecated library issues, ensuring compatibility with the latest framework
- Developed **WebSocket** reconnection for the web monitoring dashboard, ensuring robust and seamless real-time data updates

Fleetworks **San Francisco, CA**
Software Engineer Intern Jun 2024 - Oct 2024

- Engineered **AI agents** to autonomously handle carrier sales calls using GenAI, enabling automated negotiation and finalization of load and pricing details with drivers
- Designed and implemented a comprehensive, real-time carrier sales **Analytics dashboard** with over 10 distinct modules using React and Supabase; the platform provides clients with interactive data visualizations and automated alerts, enabling detailed monitoring of metrics such as call volume, handover rates, load data, and anomaly detection
- Created a tool that automated the process of sending notifications to customer companies using Webhook and **RESTful API**
- Conducted **prompt engineering** for conversation summary extraction and classification using techniques like the chain of thought, reducing the generation latency by 35% and boosting accuracy from 60% to 90%
- Established a **fully automated test framework** for evaluating prompt versions leveraging GPT-4o and o1-preview to compare accuracy and latency metrics systematically

Interactive Electronics Art Installation: *"The Living Funeral"* **Northfield, MN**
Independent Developer Sep 2023 - Nov 2023

- Extended "The Living Funeral" into a web-based project; deployed on https://www.thelivingfuneral.art using **Vercel** and **Backblaze** B2
- Developed an interactive art piece titled "The Living Funeral" using **Raspberry Pi** and **Arduino**, integrating software and hardware to create immersive visuals and scripts that stimulate reflection on existence and mortality
- Designed an end-to-end user interface with 27 different modules (e.g. **Adaptive Narration, MoodTune Panel**) to maximize the personal customization and engagement of the piece by using the Python Tkinter library
- Amplified the installation's interactivity and realism by orchestrating a customized narrative script via user inputs, leveraging the **ChatGPT API**, and providing immediate audio-visual feedback using an AI text-to-speech API
- Showcased the installation at a **campus-wide exhibition**, conducted over 50 immersive "living funeral" sessions, and garnered acclaim for the profound emotional impact and contemplative nature of the experience

RESEARCH & PUBLICATION

- Can the derivative of an inverse equal the inverse of the derivative. *Co-authored with Dr. Montee, Carleton College (Accepted by The College Mathematics Journal, https://doi.org/10.1080/07468342.2024.2392476)*
- Presented findings on balanced walls in random groups as a lead presenter at the Joint Mathematics Meetings